

Laboratory IT110

Due date: March 4th, 2026, 11:00

Bernal Manzanilla Saavedra

Lab 7: Little Man computer

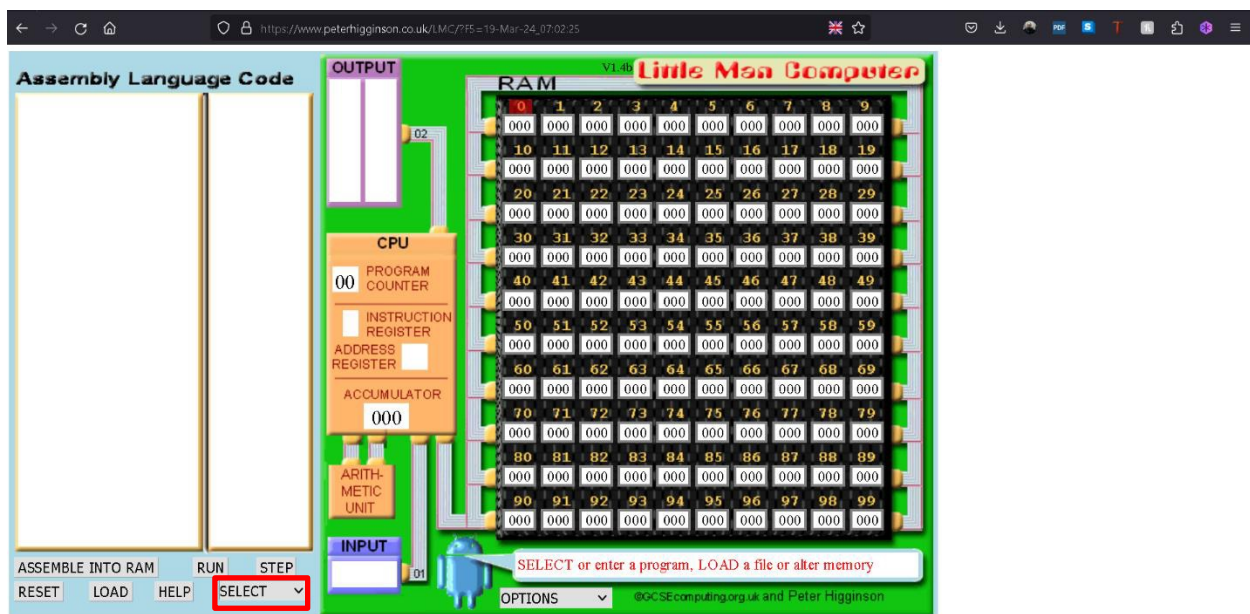
*Submit the answers to all questions for grading to Moodle. As a pdf from a word document or a scan of your handwritten answers and questions attached. Please make sure your writing is **clear and legible**. **Illegible writing might lead to deductions due to the incapacity of the instructor to assess your understanding.***

Topics:

Please access the page: <https://www.peterhigginson.co.uk/LMC>

This website is an online emulator developed by Peter Higginson. It is a pedagogical proxy, of a basic von Neumann Architecture central processing unit, with one channel for input and another for output, with no external storage that has been a favorite to teach assembler programming, developed Dr. Stuart Madnick in 1965 and called the **“Little Man Computer.”** You can read Peter Higginson here: <https://www.peterhigginson.co.uk/>.

Once you open the first link you will see the following display on your browser:



1. Click on the drop-down menu with a red square
 - a. From the dropdown menu, select “add.”
 - b. Describe what each instruction line does.

c. Describe what the program does.

2. Consider the following Little Man program:

MNEMONIC	MLBOX/OP	OP	WHERE	RESULT
Load LDA(5)	000	5	12(001)	001 → to CALC
add ADD(1)	001	1	15(390)	+ 001 = 391 → to CALC
store STO(3)	002	3	04	391
Load LDA(5)	003	5	12(001)	(001)
Halt HLT(0)	000	—	—	000
store STO(3)	004	3	91(001)	001
Load LDA(5)	005	5	12	+ 001
Add ADD(1)	006	1	13	= + 001
Store STO(3)	007	3	12	002
Load LDA(5)	008	5	14	005
Subtract SUB(2)	009	2	12(002)	003
BRANCH BZP(8) POSITIVE	010	8	00	

1. Describe what each instruction line does.
2. Describe what the program does.

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1. Click on the drop-down menu with a red square
 - a. From the dropdown menu, select “bubble sort.”
 - b. Describe what each instruction line does.
 - c. Describe what the program does. Does it have **ANY** connection to the practice Linux shell scripts seen in lab 5?
 - d. Although I do not expect that you explain EXACTLY each step I DO expect that you clearly give an explanation to what the main purpose of this program is.